Quantum-Resistant Password Management System Proposal

Overview

With the advancements in quantum computing, traditional cryptographic approaches are becoming increasingly vulnerable. To future-proof our password management system, we propose using the **Dilithium-Crystals** quantum-resistant algorithm. Below is a step-by-step outline of our proposed implementation along with the rationale for selecting Dilithium-Crystals among other NIST-recommended algorithms.

Implementation Flow

1. **Applying Quantum-Resistant Signature (Dilithium-Crystals)**
    - Upon password entry, we generate a **Dilithium-Crystals signature**, ensuring a high level of quantum security.
    - This signature, together with the **public key**, is stored on the blockchain, creating a secure reference point for future verification.
2. **Hashing with Keccak-256**
    - The password is also hashed with **Keccak-256**, a reliable algorithm widely used in blockchain, to add another layer of security.
3. **Signing with Metamask Private Key**
    - Using the user's **Metamask private key**, we sign the Keccak-256 hash, which authenticates the transaction on the Ethereum network and adds an extra verification layer.
4. **Storing Data on Blockchain**
    - The Dilithium-Crystals signature, public key, Keccak-256 hash, and Metamask signature are securely stored on the blockchain.
    - To ensure authenticity, the Metamask signature is verified using the **ecrecover** function, validating the signature against the user's address.

5. **Login Verification Process**
    - At login, the user re-enters their password, and we regenerate the Keccak-256 hash.
    - This hash is then compared against the stored data on the blockchain.
    - Additionally, a new Metamask signature is created and verified using blockchain methods to confirm the user's identity.

Why Dilithium-Crystals?

NIST recommends multiple quantum-resistant algorithms, including **SPHINCS+**, **Rainbow**, and **Dilithium-Crystals**. Here's why we selected Dilithium-Crystals:

- **Efficient Storage**: SPHINCS+ produces significantly longer signatures, which are impractical for blockchain storage. Dilithium-Crystals, in contrast, creates compact signatures that fit well within blockchain storage constraints.
- **Consistent Key Generation**: Dilithium-Crystals is unique in offering reliable, consistent keys that allow us to store a single key on-chain and use it for verification at later stages.
- **Resilient Security**: With strong resilience against both classical and quantum attacks, Dilithium-Crystals provides robust security guarantees essential for this system.

Multi-Layered Security

To maximize security, we are implementing **three distinct layers** of protection:

- **Front-End Quantum-Resistant Signature**: Dilithium-Crystals applied on the front end ensures protection against quantum threats.
- **Backend Keccak Hash**: The Keccak-256 hash provides backend-level protection, safeguarding the password on-chain.

- **Frontend-Backend Connection Verification**: By requiring Metamask signing, we ensure that only verified front-end applications interact with the blockchain, preventing unauthorized direct access.

This approach allows us to achieve high levels of security, integrating quantum resistance and conventional blockchain protections, and delivering a robust system that’s well-prepared for current and future threats.